



Retail Supply Chain Analytics

Project Report

Executive Summary

An end-to-end analytical and machine learning study completed for the **NTI Data Analytics Track Final Project**. The project focuses on predicting **Daily Demand** and evaluating supply chain risk metrics across operational, logistics, and financial dimensions using Python, Scikit-Learn, and Power BI.

1. Dataset & Data Quality

- **Scale & Focus:** 44 curated transaction records focused on **Daily Demand Detection**.
- **Data Quality:** 100% Clean — **0 missing values** and **0 duplicate rows**.
- **Scope:** Covers operational stock, lead/delivery times, transport modes, and pricing metrics.
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2. Machine Learning Performance (Daily Demand Detection)

- **Algorithm:** GradientBoostingRegressor
- **Target Variable:** Daily Demand (Daily_Demand)

Performance Metrics:

- **MAE:** 24.89 units
- **RMSE:** 28.78 units
- **R² Score:** -0.001

Technical Diagnostic Note: The baseline R² approx 0 indicates that static snapshot data (44 records) lacks historical sequence signals. It establishes a clear technical baseline proving that sequential **time-series logging** is required for accurate automated demand forecasting.

3. Power BI Dashboard Structure

A 4-page interactive dashboard suite covering:

1. **Product & Demand Overview:** Demand distribution and stockout risk.
2. **Warehouse Operations:** Capacity, inventory throughput, and reorder levels.
3. **Logistics & Freight:** Delivery transit times and transport mode efficiency (Air, Sea, Road).
4. **Supplier Performance:** Vendor reliability scores, lead times, and SLA compliance.

4. Key Strategic Recommendations

- **Implement Time-Series Infrastructure:** Shift to sequential time-series models (e.g., ARIMA/Prophet) to capture seasonal demand trends.
- **Safety Stock Buffers:** Use the MAE margin (+_ 25 units) to adjust safety stock thresholds and prevent stockouts.
- **Logistics Optimization:** Re-evaluate carrier SLAs and prioritize transport modes based on margin impact and delivery windows.

5. Project Information

- **Track:** NTI Data Analytics Track — Final Graduation Project
- **Core Tech Stack:** Python (Pandas, NumPy, Scikit-Learn), Power BI, GitHub